Appendix L. Bochs

Date: 2026-10-20

The command card for the host that opened the lesson. Chapter 7.6 is the claim. Chapter 4.6 is what the stops showed. `maps/bochs-20261020.md` is that boot's selectors. This appendix is how to do it again.

L.1 Why this host

Bochs runs the Win98 guest and holds the debugger outside it. A break does not need SoftICE, and it does not need a program inside Windows. Registers, the stack, and memory are what the CPU holds. The cost is speed.

L.2 This machine

Item	Value
Build	`~/bochs/bochs/build-dbg/bochs` (3.1, internal debugger). `build-rfb` is the VNC build and was not used.
Config	`~/bochs/vm/win98.bxrc`
Disk	`~/bochs/vm/win98.img` (2 GB flat, cylinders=4063, heads=16, spt=63)
Guest	Win98 SE, 128 MB, Pentium model, ips=50000000, i440fx, Cirrus VGA, PS/2 mouse
BIOS	`~/bochs/bochs/bios/BIOS-bochs-latest` and `/usr/share/bochs/VGABIOS-lgpl-latest`
Dumps	`~/bochs/vm/dump*/`

Start:

```
~/bochs/bochs/build-dbg/bochs -f ~/bochs/vm/win98.bxrc -q -unlock -debugger -dbglog ~/bochs/vm/debug.log
```

`-debugger` opens `<bochs:N>` in the terminal. `c` runs the guest. Ctrl+C returns to the prompt. `-unlock` clears a stale lock left by a killed Bochs. `-dbglog` saves what the debugger prints, including a long `info ldt`.

The guest window and the terminal are separate. Click the window for the mouse and the keyboard. Ctrl+F10 toggles the mouse. Read the terminal before the window. A missing BIOS is a panic there, not a Windows error.

L.3 Files into the guest

```
cd ~/<folder>
genisoimage -o ~/files.iso -J -R .
```

Point `ata0-slave` in `win98.bxrc` at that ISO with `status=inserted`, then restart. Copying a file back out of the guest has not been needed. The `genisoimage` flags were not re-tested this session.

L.4 Commands that worked

Command	What it does
`vb 0xSEL:0xOFF`	Break at a live selector:offset.
`info break`	List breakpoints. The numbers are what `bpd` and `bpe` use.
`bpd N` / `bpe N`	Disable or enable breakpoint N.
`c`	Continue. One command per line.
`r`	Registers. A 16-bit read is the low half: AX, DX, not EAX.
`print-stack N`	N stack words. Addresses are linear.
`info ldt A B`	LDT entries by decimal index. `2333 2333`, not `0x48ef>>3`.
`x /20xb ADDR`	Bytes at a linear address. Not selector:offset.
`writemem "file" ADDR LEN`	Save guest memory. The folder must already exist. Stops at the first unmapped page.
`source file`	Run a command file. Absolute path.

Not used, so not trusted: `del N`, `lb`, `trace on`, `show call`, `loadmem`. There is no debugger command for a full physical RAM dump. The GUI Suspend hung. Do not use it. Kill with `kill $(pgrep -n bochs)` and start again with `-unlock`.

L.5 Reading a stop

Bochs prints `Breakpoint K`, the selector:offset, and the instruction.

At a function entry the first two stack words are the return address: offset, then segment. The arguments follow, nearest the top first. That order matches a Ghidra `__stdcall16far` signature.

```
print-stack 24
  STACK 0x80477ac0 [0x0a60]   return offset
  STACK 0x80477ac2 [0x38f7]   return segment (11B0 this boot)
  STACK 0x80477ac4 [0x0c36]   first argument
```

A far pointer comes back in DX:AX. A word comes back in AX. To catch the return, break on the instruction after the call. `28FF:02E5` was the instruction after the `03C8` lookup on this boot.

L.6 Selectors after a reboot

Windows assigns an LDT selector when the NE segment is loaded. The numbers change every boot. Only a segment that has been touched has a real entry.

1. Start with `-dbglog`. Open the lesson so TB80BAS loads.
2. `info ldt`. Match Ghidra segments by size, not by the index formula.
3. Confirm with a relocation. A far call in the file has the Ghidra selector (`0x10B0` and the rest). The same word in the loaded bytes has the live selector.

The table for the boot of 2026-10-20 is `maps/bochs-20261020.md`. Do not reuse it.

L.7 One object load

Set these only after the lesson has been opened once, so the segments exist. Then open the page. The selectors are the 20 October boot.

```
vb 0x2a37:0x2174   book open
vb 0x2907:0x08b2   reader creator     handle, 0x096c, err
vb 0x391f:0x04da   entry read         reader far, 2, entry far, slot, err
vb 0x391f:0x08d6   read setup
vb 0x291f:0x02cc   file read          len, dest off, dest seg, pos lo, pos hi, file, err
vb 0x28ff:0x02ce   send message       0, 0, msg, class word, block, handle, err
vb 0x28ff:0x03c8   handler lookup
vb 0x28ff:0x02e5   lookup returned    DX:AX is the handler
vb 0x2907:0x0aeb   after message 0xE
```

Every object hits several of these. `bpd N` silences a stop that is already understood. The handle (`0x0C1E`, `0x0C36`, `0x0C4E`) names the object. Those three are `0x18` apart.

To dump the record: at `08D6`, note the block selector. After `02CC` returns, `info ldt N N` for that selector's decimal index, then `writemem` the base for `limit + 1` bytes.

The on-disk copy of entry index `I` is at file `0x10D6 + 24 * I`.

L.8 Pitfalls

What happened	What to do
Bochs will not start. Files locked.	Add `-unlock`.
Panic reading a ROM image.	The BIOS path in the config is wrong. Read the terminal.
Guest window stays small.	This X build does not scale. Not solved.
Mouse stuck on the zoom tool.	That was the desktop magnifier. Click the guest. Ctrl+F10.
`syntax error at print-stack`	More than one command was pasted. One line at a time.
`writemem` writes nothing.	The folder is missing, or the address is unmapped. `mkdir` first. `info ldt` first.
Suspend hangs.	Do not use it. Kill Bochs. Restart with `-unlock`.
The dump is the wrong segment.	The first data dump used LDT index 1161. The data segment was `3477` this boot.
The bytes look right and the selector is wrong.	Selectors changed. Section L.6.

L.9 Still to capture

Each of these needs the lesson opened again, so the selector map comes first.

1 The dispatcher `03C8` returns for handle `0x0C36`. The return already captured was handle `0x0C1E`, which is not the 412-byte record.

2 Twenty-four bytes of the live entry. On this boot that was `x /24xb` at the `3E67` base plus `0x0C36`. The file's last four bytes are `00 21 25 00`. They have not been seen in memory.

3 The file record at slot `0x096C` and file-table offset `0x0AE0`, with the book open, including the class-table pointer at `+0x163`.

4 The class table that pointer names.